

Unhittable

WCAG says a button must be **24 pixels**. We replayed **52 clinically recorded tremors** against that number, then measured **3,842 real controls** on 38 live sites.

THE RULE THAT GOVERNS EVERY BUTTON YOU SHIP

WCAG 2.2, SC 2.5.8, Level AA.

Every accessibility checker on the market measures your controls against **24 by 24 CSS pixels** and prints a green tick. axe-core, Lighthouse, whatever your company pays for.

Its published basis is **Parhi, Karlson & Bederson, MobileHCI 2006**: a one-handed thumb study of **able-bodied participants** on small touchscreens.

axe-core's own rule page names "people who have mobility impairments such as hand tremors" as the beneficiaries, and then applies 24 px to everyone.

WHAT WE MEASURED

Four steps, all real.

1· Clinical recordings

PADS on PhysioNet · Apple Watch · 100 Hz · 260
people

2· Acceleration to distance

band-limited double integration, gyro-corrected

3· Distance to pixels

800 cpi mouse: 1 mm of hand = 31.5 CSS px

4· Against a real page

headless Chromium measures every control as
rendered

target size is a LAYOUT property, so the page must actually be laid out

THE COHORT, NOT ONE FLATTERING RECORDING

1,560 recordings. 260 assessed people.

Only **52** contain a tremor that clears a bar **zero of 2,300 null draws** reach. Most recordings have none, and we publish that.

1.88 mm

median amplitude of the 52, which is 59 px of cursor travel

46

of 52 are wider than the entire 24 px target

36

hold a 24 px target less than 95% of the time

25

still fail at the 44 px AAA enhanced size

essential tremor has the highest detection rate, 14.6%, and affects ~10x more people than Parkinson's

THEN WE POINTED IT AT THE WEB

38 live sites. 3,842 controls.

Government services, healthcare, the disease organisations whose own audiences have tremor, and the accessibility vendors who sell the audits.

95.9%

PASS WCAG 2.5.8, including the spacing exception

72.1%

of those cannot be reliably held by a median recorded tremor

142×29

the median control on the web, in CSS pixels

WHY A SQUARE MINIMUM CANNOT SEE IT

The web is built from wide, short buttons.

A 141 by 30 control passes, because both sides clear 24. The **short side** is what a hand actually has to hold, and a single number for "target size" hides which one it is.

On the Social Security Administration's homepage: **108 controls, zero fail the standard, 95 pass it and still cannot be held.**

A finding, and the instrument that produced it.

WEB

Paste any URL. Real Chromium loads it, waits for fonts, measures every control as rendered, and replays the recording against each rectangle. About three seconds.

Everything on the page is recomputed in your browser from the committed recordings. Nothing is precomputed.

ANDROID

Measures **your** hand from the phone's accelerometer, works out the button size you need, then **checks that prediction against your own taps.**

On a touchscreen a millimetre of finger is a millimetre off target, with no counts-per-inch to sweep. Calibrate against a bank card and the millimetres are real.

WHAT WE GOT WRONG, AND HOW WE KNOW

An adversarial review nearly killed this.

A Hann window never removed

a true 2.000 mm read as 0.778 mm at the record's edge

Then: we fixed a confound the data lacks

the filter assumed gravity; PADS ships none, and reported 73° of tilt on a still wrist

A robots.txt matcher

silently excluded 18 of 40 sites, all wrongly

DNS rebinding in the scanner

two lookups, and an attacker answers them differently

The second was found in the round that reviewed the fix for the first. A synthetic test is made out of your assumption, so it can never tell you the assumption is false.

fourteen of these are written up in docs/FALSE-GREENS.md

WHAT WE ARE NOT CLAIMING

The honest boundary.

We do not diagnose anyone

condition labels are the dataset's, verbatim

Measuring tremor is solved

StudyMyTremor has shipped millimetres since 2016

Adaptive sizing is not new

SUPPLE++ 2007, Sarcar 2018, ABD-MT 2024

Most recordings show nothing

52 of 1,560, and we publish the other 1,508

Our own page fails too

44 px is AAA and still not enough for this hand

New here: clinically recorded tremor, in physical units, evaluated against WCAG 2.5.8 itself, on pages that are actually shipping.

WHO IT IS FOR

About **11 million** hands in the US alone.

Roughly **10 million** people live with essential tremor and **1 million** with Parkinson's. Every one of them is currently told by an automated green check that your 24-pixel button is fine.

The fix is not a new standard. It is a number that was never checked, and a tool that checks it in three seconds against a page you already shipped.